MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE: M37

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of squib actuated switch assembly designated as Semiconductor Device Set: Training, M37 which is used with the Honest John warhead section.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8833509	- Semiconductor Device Set: Training, M37.
EL-8822509	- Equipment List.

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The semiconductor device set shall comply with all requirements specified on the drawing (dwg) 8833509, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Contact resistance. The electrical resistance measured between pin pairs of the electrical receptacle shall not exceed 0.030 ohms when tested as specified in 4.4.1.

3.5 Insulation resistance. The insulation resistance between pins and between all pins and the case (except between those electrically connected) shall be not less then 20 megohms when tested as specified in 4.4.2.

3.6 Environmental testing. The semiconductor device set shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The semiconductor device set shall also comply with 3.4 and 3.5 before and after each environmental test. Tests shall be conducted in the order specified.

3.6.1 Humidity. The semiconductor device set shall withstand, without damage, moisture conditions as follows:

> Exposure to a temperature of 120 degrees Fahrenheit (°F) and 95 percent relative humidity for a period of 15 days.

The test shall be as specified in 4.4.3.1.

3.7 Soldering. All soldering shall be in accordance with MIL-STD-1460.

3.8 Workmanship.

3.8.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.8.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.8.3 Burrs. Parts shall be free from burrs.

3.8.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

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3.8.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 8 complete assemblies (see dwg 8833509) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table I. Tests shall be conducted in order specified in Table I.

Table I
First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Contact resistance (see 3.4)		X
3. Insulation resistance (see 3.5)		X
4. Humidity (see 3.6.1)	X	

4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

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4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table II and Table III), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table II
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table III
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Semiconductor device set assembly, prior to potting (see dwg 8833509).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Insulation inadequate or damaged	Visual
	102. Foreign matter	Visual
	103. Any component loose, damaged or missing (terminals, fuses, resistors, diodes, connectors, connector pins, or brackets)	Visual/Manual
	104. Solder loose or insufficient (see 3.7)	Visual/Manual
	105. Solder beads loose (see 3.7)	Visual/Manual
	106. Objective evidence relating to material missing or improper (see 3.1)	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.3.2.1.2 Semiconductor device set assembly (see dwg 8833509).

Categories	Characteristics	Methof of Inspection
Critical:	None defined.	
Major:		
	101. Location of connector and mounting holes	Gage
	102. Protective coating exposing bare metal	Visual

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103. Any operation incomplete or missing.....Visual
104. Foreign matter.....Visual
105. Potting inadequate.....Visual
106. Pins in connector damaged, loose or missing.....Visual/Manual
107. Hardware loose, damaged or missing.....Visual/Manual
108. Any component loose, damaged or missing.....Visual/Manual
109. Objective evidence relating to material missing or improper (see 3.1).....Visual

Minor:
201. Marking on identification plate improper, illegible or missing.....Visual
202. Evidence of poor workmanship (see 3.8).....Visual

4.3.2.1.3 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
	101. Preservation packing and packaging not in accordance with MIL-STD-1169	Visual
	102. Marking not in accordance with MIL-STD-1169	Visual
Minor:		
	201. Evidence of poor workmanship (see 3.8)	Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test classification
a. Contact resistance	Major
b. Insulation resistance	Major

None of these tests are considered destructive and samples so tested may be returned to the lot.

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4.3.3.1 Contact resistance and insulation resistance. Each semiconductor device set assembly in the lot shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.4 and 3.5, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8833509). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Contact resistance. The electrical resistance measured between the following pin pairs of the electrical receptacle shall not exceed the specified resistance:

1 to 2	11 to 29	21 to 22
1 to 11	13 to 14	26 to 27
3 to 21	13 to 31	27 to 28
6 to 25	14 to 15	29 to 30
7 to 8	17 to 35	31 to 32
7 to 26	18 to 19	32 to 33
8 to 9	19 to 37	35 to 36

4.4.2 Insulation resistance. The semiconductor device set shall be subjected to a d.c. potential of 500 ± 50 volts until a steady static reading is obtained using a megohm meter from which a direct reading of resistance in megohms can be made.

4.4.3 Environmental testing (see 3.6).

4.4.3.1 Humidity (see 3.6.1). The semiconductor device set shall be tested in accordance with MIL-STD-810, Method 507, Procedure I except that the maximum temperature shall be 120°F and the number of cycles shall be 15. Within 2 hours after removal from the humidity chamber the assembly shall be tested as specified in 4.4.1 and 4.4.2.

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5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The semiconductor device set shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Level C. The semiconductor device set shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.2 Packing.

5.2.1 Levels A and C. The semiconductor device set shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-N-I.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1806 dated 16 August 1960.

Custodian:
Army - PA

Preparing Activity:
Army - PA

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